AI Lab: Decision Trees & Random Forest

BS DS FALL 2024

Lab # 04

TASK 01: Decision Tree Classifier

Introduction to Decision Tree:

Decision Tree is a supervised learning algorithm used for classification problems.
It splits data into branches based on feature values.

Key Points:
- Works like a flowchart
- Uses conditions to split data
- Can overfit if depth is too large

Code Template:

```
class DecisionTreeModel:

    def __init__(self, max_depth=5):
        self.max_depth = max_depth

    def load_data(self):
        # Load dataset
        pass

    def preprocess(self, df):
        # Handle missing values and encode categorical data
        pass

    def train(self, X_train, y_train):
        # Train Decision Tree model
        pass

    def predict(self, X_test):
        # Predict output
        pass
```

```
    def evaluate(self, y_test, y_pred):
        # Calculate accuracy, precision, recall
        pass
```

Lab Tasks:

1. Implement Decision Tree for Loan Approval dataset.
2. Run model with different depths:
    - Depth = 2
    - Depth = 5
    - Depth = None
3. Students must record:
    - Accuracy
    - Precision
    - Recall
4. Compare how depth affects:
    - Over-fitting
    - Performance

TASK 02: Random Forest

Introduction to Random Forest:

Random Forest is an ensemble learning method.
It combines multiple decision trees to improve accuracy and reduce overfitting.

Key Points:
- Uses multiple trees
- Uses random sampling
- More stable than Decision Tree

Code Template:

```
class RandomForestModel:

    def __init__(self, n_estimators=100):
        self.n_estimators = n_estimators

    def train(self, X_train, y_train):
        # Train Random Forest model
        pass
```

```
    def predict(self, X_test):
        # Predict output
        pass

    def evaluate(self, y_test, y_pred):
        # Calculate accuracy, precision, recall
        pass
```

Lab Tasks:

1. Implement Random Forest for Loan Approval dataset.
2. Run model with:
    - n_estimators = 10
    - n_estimators = 50
    - n_estimators = 100

3. Students must record:
    - Accuracy
    - Precision
    - Recall

4. Compare performance with Decision Tree.

Main Function and Running Example:

```
def main():
    # Load dataset
    # Preprocess data
    # Split into train/test
    # Train Decision Tree and Random Forest
    # Print results
    pass

main()
```